3 Information and Communication Technologies in Network Services

Network

- A network is an interconnection of a series of ports, or nodes, connected by a communication path. Networks can interconnect with other networks and may contain subnetworks.
- To connect the nodes, a media/channel is required, which is known as transmission media. A simple network consists of a transmitter and a receiver separated by a distance and interconnected by transmission media.
- A sender node can be anything like an antenna or a computer. If two computers are interconnected by a transmission media, it corresponds to a computer network.

Transmission Media

- Media is the general term used to describe the data path that forms the physical channel between sender and receiver. Transmission media can be twisted pair wires such as that used for telephone installation, wired media are referred to as bounded media and wireless media are sometimes referred to as unbounded media.
- Different types of transmission media are used for different data transfer rates and long distances. Bandwidth, noise, radiation, and attenuation are considered while using the different transmission media.
- Higher bandwidth transmission media supports higher data rates.
- Attenuation limits the usable distance that data can travel on the media. Noise is related to an electrical signal that can distort data signals and data errors.
- Radiation is the leakage of signal from the media caused by undesirable electrical characteristics of the transmission media.
- The transmission medium is the physical path between the transmitter and receiver in a data transmission system. Transmission is determined both by the characteristics of the medium and the characteristics of the signal.

Classification of Transmission Media

The transmission medium can be mainly classified into the following two types

- Bounded or guided media
- Unbounded or unguided media

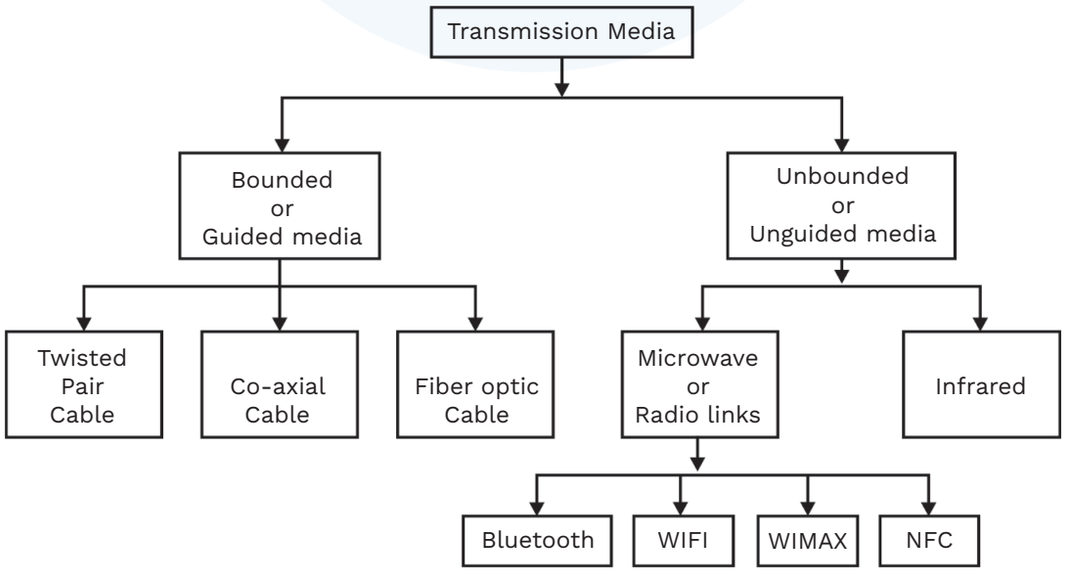


Fig. 3.1

Bounded or Guided Media

In bounded media, the wave propagation is through a fixed guided path. The bounded media is also called wired media. Depending on the type of transmission medium used, the bounded media can be further classified into three types.

- Twisted pair cable
- Coaxial cable
- Fibre optic cable

Unbounded or Unguided Media

Depending on the method of transmission, the unbounded media can be further classified into the following two types

- Microwave links or radio links
- Infrared

Wi-Fi or Wireless Fidelity

- Utilising radio waves and a computer network, such as the Internet, Wi-Fi enables electronic devices to communicate with one another. The Wi-Fi brand name and logo are the registered trademarks of the Wi-Fi Association, which promotes the IEEE 802.11 series of wireless networking technologies.
- Access points (sometimes known as 'hotspots') have a larger range outside, but still have a usable range of roughly 20 m indoors. Hotspot penetration may range from covering an area the size of a single room to hundreds of square miles. Several overlapping network nodes are used to accomplish this.
- Wireless network access points allow Wi-Fi enabled devices, such as computers, gaming consoles, smartphones, tablets, and digital audio players, to gain access to networks and their associated resources, such as the Internet.
- Using Wi-Fi, local area network (LAN) implementation costs may be reduced. Wireless local area networks (WLANs) may be set up in places where it would be impractical to lay cables, such as in the great outdoors or in old structures. If a robust passphrase is used, Wi-Fi protected access 2 (WPA2) encryption is secure.
- This device is dual band, functioning at both the 2.4 and 5 GHz ranges. The range of 2.4 GHz Wi-Fi networks is somewhat greater than those operating in the 5 GHz band.

Bluetooth

- Frequency hopping spread spectrum is the radio technology used by bluetooth, and it splits the information being transferred into smaller pieces before sending them in bursts over a band of frequencies, from 2400 to 2483.5 MHz, in increments of 1 MHz. With bluetooth, you can set up secure personal area networks (PANs) between stationary and mobile devices that can exchange small amounts of data.
- There is no need for a direct line of sight between bluetooth devices since they communicate through radio (broadcast) technology. Bluetooth allows for the safe pairing and transfer of data between a wide variety of electronic gadgets, including but not limited to mobile phones, fax machines, laptops, landline telephones, desktop PCs, printers, GPS receivers, and gaming consoles.

Bluetooth vs Wi-Fi

- Applications such as network setup, printing, and file transfer are identical across bluetooth and Wi-Fi.
- As its name suggests, bluetooth was designed to work with mobile devices and their related software. Wireless personal area networks are described as a category of applications (WPAN). Wi-Fi was designed to be used instead of hardwired LAN connections in commercial and institutional settings. Wireless local area networks (WLAN) is a term occasionally used to describe this class of applications.
- Bluetooth may be used instead of wires in a wide range of portable applications, and it can also be used in stationary ones, such as smart electricity management in the house (thermostats, etc.).
- As the wireless equivalent of the standard wired Ethernet network, Wi-Fi needs to be set up to set up shared resources, transfer files,

and set up audio connections, for example, headsets. Wi-Fi employs the same wireless signals as bluetooth, except with more power, allowing for greater data transfer rates and greater distances from the access point.

Worldwide Interoperability for Microwave Access

- In the context of wireless networking, Worldwide Interoperability for Microwave Access (WiMAX) is considered to be a 4G technology. WiMax offers a metropolitan region connection with a broadcast radius of roughly 50 km, far further than the communication range of a normal Wi-Fi LAN, which is typically about 30 m.
- Fixed WiMax, formerly known as IEEE 802.16, is a mobile communications standard that was updated to provide data speeds of up to 1 GB/s at fixed stations in 2011. While theoretically, WiMax's data transfer rates may be higher than those of standard cable modem and digital subscriber line (DSL) connections, in reality, those theoretical benefits would be tempered by the need to share the available bandwidth among several users.
- Although there is not a single, standardised licenced spectrum available worldwide for WiMAX, the WiMAX Group has released three licenced spectrum profiles (at 2.3, 2.5, and 3.5 GHz) in an attempt to promote standardisation and bring down costs.
- WiMAX's bandwidth and range make it a promising technology for a number of uses, including.
    - Delivering mobile broadband access on the go across cities and nations with a wide range of devices.
    - As part of a disaster recovery strategy, providing Internet access is a must. The ability to stream media via the internet, as well as voice over IP (VolP) telephone service.

Wi-Fi vs WiMAX

- More consumer electronics these days make use of Wi-Fi, which utilises unlicenced airwaves to connect to a local area network. WiMAX is a long range technology that may travel several kilometres and links users to a network (often the Internet) through licenced or unlicenced airwaves.
- Wi-Fi relies on contention access, which means that all subscriber stations trying to transmit data via an AP must vie for the AP's attention on an interrupt by interrupt basis. This might drastically decrease the throughput for subscriber stations that are farther away from the AP due to frequent interruptions from stations that are closer to the AP. WiMAX, on the other hand, employs a QoS system predicated on links between the base station and the user equipment. Different scheduling methods underpin various connections.
- WiMAX Subscriber Units are often supplied by WiMAX network operators, connecting local Wi-Fi enabled devices (such as laptops, Wi-Fi handsets, and smartphones) to the metropolitan WiMAX network and enabling them to access the internet from inside a building. Because of this, the user may position the WiMAX Subscriber Unit in the best reception location (such as a window) while still enjoying full WiMAX network access from wherever in the home. Wi-Fi and WiMAX serve distinct purposes, yet they complement one another.

Near Field Communication (NFC)

- Near Field Communication (NFC) is a collection of protocols that allow cell phones and similar technologies to initiate radio transmission with each other simply by contacting or coming within a few centimetres of each other. Whereas previous systems, such as contactless smart cards, only allowed for one-way communication between terminals, NFC expands upon RFID (radio frequency identification) technologies by allowing for two-way communication.
- Contactless payments, data transfers, and streamlined configuration of sophisticated networks like Wi-Fi are just some of the current and future uses.
- Rural financial services, ticket booking services, interactive and focused marketing, leisure, and pharmacies are only a few examples of where NFC has found a home.

- A credit card might be replaced by a smartphone with near field communication capabilities. The reason for this is that NFC enabled phones have a 'smart chip' with a very difficult to break 80 character code. A gadget such as this would be ideal for stowing away credit card information securely while still being convenient for purchasing items on the fly.
- You may utilise NFC to interact with other people in a variety of social contexts, such as exchanging contact information, sharing media, and playing mobile games with other players. Your NFC smartphone may soon be the only key you need to unlock your house, your vehicle, and your workplace.

Comparison between NFC and Bluetooth

- Bluetooth, like near-field communication (NFC), is a short range technology, but it enables engagement between telecommunication devices up to 10 m away, while NFC only enables contact between devices within a few centimetres of each other. Although NFC is slower than bluetooth, it uses much less energy and does not necessitate pairing.
- The initial setup for NFC is quicker than for bluetooth 1.0, but it is slower than for bluetooth low energy. Compared to bluetooth V2.1's 2.1 Mbps, NFC's highest data transmission rate of 424 Kbps is rather modest. Instead of manually configuring devices, a link may be created in less than a tenth of a second using NFC.

Computer Network

- A network is a combination of two or more computers connected for the exchange of information. The computers connected are independent i.e., they can independently start and stop on control on their own.
- A network allows computers to share their equipment, programmes, and information available to each other. To interconnect the different computers placed at different places, communication channels such as cable, telephone lines, or microwaves are used.

Goal of the Networking

- The primary motivation for networking is 'resource sharing,' which means that all resources (including but not limited to data, applications, and hardware) should be accessible to all users on the network regardless of their location.
- To save costs is another priority. In terms of cost and functionality, compact computers excel above their bigger counterparts. Such an objective motivates the deployment of networks in which several machines are housed in the same physical location. This kind of network is known as a local area network (LAN).
- Having several redundant supply routes is another need with this design. If a server goes down, the other copies of the files will still be accessible.
- Computer networks are an efficient means of communication. Any changes made to a file on a shared network are instantly visible to all other users of that shared network.
- A second, related objective is to scale up the system's processing power in step with the growth of the workload. When a central mainframe reaches capacity, it should be changed to a newer, bigger system, which is a time-consuming, expensive process that often causes interruption for the users.
- Networking in computers is used to
  - Share costly devices, such as printers.
  - Communicate with each other.
  - Improve communication speed and accuracy.
  - Reduce the cost of data transfer.
- The various components of a network are described below.

Media

It is the media that provides a physical connection. It can be anything like twisted pair cable, coaxial cable, fibre optics, microwave radio, satellite cellular systems, etc.

Processors

These are the devices that support network connectivity and transmission. It includes various devices such as modems, multiplexers, switches, routers, etc.

Modems

Modems stand for modulator + demodulator. During transmission, the message signal is modulated with a carrier signal to convert it to an appropriate signal for transmission, which is then demodulated to recover the original message signal.

Multiplexers

It is a device that is used to combine data from multiple users to transmit it over a common path.

Switch

It is a communication processor that is used to make the connection between various nodes of a network so that a signal can reach its destination.

Router

It is an intelligent communication processor that makes the connection between various nodes of a processor with the help of protocols or different rules so that a signal can reach its destination.

Hub

It is a port switching communication processor, which is used for automatic switching of various parts and shared access to any network's access. Workstations, servers, printers, and other resources are connected to ports by the hub of other networks.

Gateways

- It is the communication processor, which connects the network and uses different communication architectures.
- We can further classify a network as
  - Based on network topology
  - Based on network extent

Based on Network Topology

The network topology defines the physical interconnection of its constituent elements. We can say the topology of a network refers to the way nodes (or hosts) are interconnected in a network. The interconnection may be either real or logical. Real interconnection refers to the actual (i.e., physical) connection of a network, while logical interconnection refers to the way data is exchanged between constituent elements. The network topology can be classified as follows.

Bus Topology

- Bus topology is used when a network installation is small. In the case of a bus network, the various nodes, such as a computer, as shown below in the figure, are interconnected through the cable. A cable is one or more wires with no active electronics to amplify the signal i.e., a passive bus topology.
- When one computer sends a signal up to the cable, all the computers on the network receive the information, but the one with the address that matches the one encoded in the message accepts the information while all others reject the message. In this scheme, the speed is slow as at a time only one computer can send a message on the bus. A computer must wait until the bus is free before it can transmit.
- The bus topology requires proper termination at both ends of the cable. Without termination, when the signal reaches the end of the cable, it returns and travels to break up the cable. The transmitted wave and the reflected wave, if they are in phase, they add each other otherwise cancel each other. Thus, to prevent the generation of a standing wave, which may distort the signal, an impedance equal to characteristic impedance is attached at the cable end.
- The bus topology is easy to understand, install, and use for small networks. The cabling cost is less as the bus topology requires the least amount of cable to connect the computers.
- The bus topology is easy to expand by joining two cables with a Bayonet Neill–Concelman (BNC) connector.
- In the expansion of a bust topology, repeaters can be used to boost the signal and increase the distance.
- Heavy network traffic slows down the bus speed. In a bus topology, only one computer can transmit at a time, and others have to wait for their turn. There is no coordination between the computers for receiving the transmitting time slot.
- The BNC connectors for the expansion of the bus attenuate the signal considerably.

- A cable break or loose BNC connector causes reflection and brings down the whole network, causing all network activity to stop.

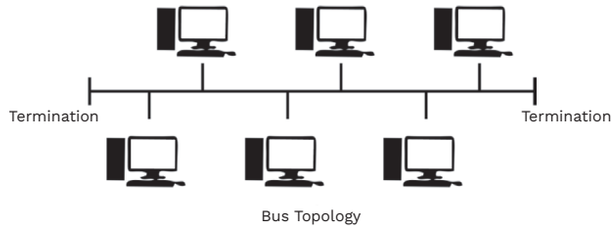


Fig. 3.2

Ring Topology

- In a ring topology, each computer is connected to the next computer, with the last one being connected to the first. Rings are used in a high-performance network where large bandwidth is necessary. The computer is connected to the next peer on the ring, and each retransmits and receives from the previous computer.
- The message flows around the ring in one direction. There is no termination of the ring because there is no end. Some rings do token passing. A short message called a token is passed around the computers in the ring from one to another. The computer that transmits the data keeps the token with it, otherwise it passes the token to the computer, which is next to it in the ring. After transmitting the data, the token is sent to the next computer in the ring. Thus, the token circulates until a station is ready to send and capture the token.
- No one computer can monopolise the network because every computer is given equal access to the token.
- The fair sharing of the network allows the network to continue to function in a useful manner.
- The failure of one computer on the ring can affect the whole network. The initial installation cost is high and hence not preferable for low density traffic.
- If any link breaks or if any repeater fails, then the entire network will be disabled.
- To install a new repeater to support a new device, it is necessary to have the identification of two nearby networks.
- The closed nature of the ring topology makes it necessary to remove the circulating packets.

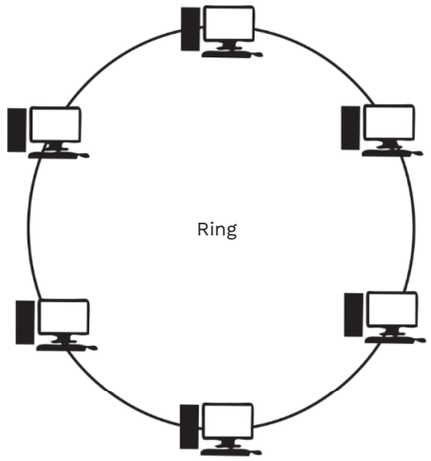


Fig. 3.3

Star Topology

- In a star topology, all computers run from the computers to a central location where they are all connected by a device called a hub as shown figure.

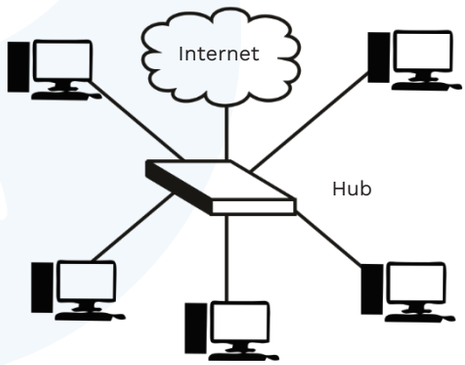


Fig. 3.4

- Each node/computer on a star network communicates with a central hub that resends the message to all computers in a broadcast star network or only to the destination computer in a switched star network. The hub in a broadcast star can be active or passive. An active hub generates the electrical signal and sends it to all computers connected to it. Active hubs need a power supply, while passive hubs do not.
- As compared to the bus topology, it gives much better performance; signals do not necessarily get transmitted to all the workstations.

- It is easy to connect new nodes or devices. In the star topology, new nodes can be added/ removed easily without affecting the rest of the network.
- Failure of one node or link does not affect the rest of the network. At the same time, it is easy to detect the failure and troubleshoot it.
- There is a huge dependence on the central hub. If it fails, the whole network fails to operate.
- Many star networks require a device at the central point to rebroadcast or switch the network traffic.
- Cabling costs are higher since cables are connected to all nodes/computers from a central hub.

Mesh Topology

- In case of the mesh topology, there is a dedicated point to point link from one device to another as shown below.

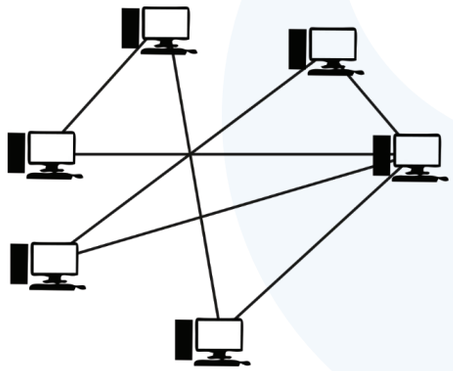


Fig. 3.5

- If there are N devices, then a fully connected network will have physical channels to link devices. Each device in the network contains (n–1) input/output ports. The links between the devices in this case are dedicated links i.e., traffic is carried on it between two devices.
- The use of a dedicated link guarantees that each connection can carry its data load, thus eliminating traffic problems.
- In the case of mesh topology, the failure of one node/computer does not bring down the entire network.
- It provides security and privacy because every message is sent along a dedicated line. In the case of failure, troubleshooting is easy.
- Installation and reconfiguration of the mesh topology are difficult since all nodes/ computers are connected node/computer.
- Cabling costs are higher. The hardware required to connect each link input/output and cable is expensive.

Based on Network Extent

Network topology is tightly coupled with network extent, and this classification leads to different types of area network. This is needed because networks come in different shapes and sizes. In addition, the extent of a network has a significant influence on its behaviour. Network can be classifiedinto various types based on network extent as local area network, metropolitan area network, wide area network, personal area network, and internet.

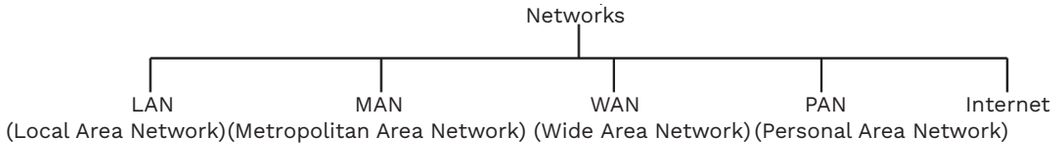


Fig. 3.6

Local Area Network (LAN)

- A local area network (LAN) is a network restricted to a small region. LANs are widely used in business enterprises to connect personal computers, network servers, printers, and other network entities. The main characteristics of LAN are
  - They have diameters of the order of a few kilometres.
  - The main aim of LAN is to provide access to data terminal equipment such as printers and plotters within the office.
  - A LAN is composed of computer hardware transmission media and software.

- LAN has high total bandwidth, so high data rates (10 Mbps) are possible on LAN.
- The delay is small and the different stations have peer relationship.
- While local area networks (LANs) may transfer data at speeds far above those of standard telephone lines, these networks are restricted in range and in the total number of systems that can be linked together.
- The most popular kinds of local area networks are outlined in IEEE standard 802.
- Point to point communication between stations via a shared physical medium is made possible by LANs because they eliminate the need for a central switching hub.

- The main characteristics that differentiate one LAN from another are
  - Topology such as star, bus, ring, etc.
  - Protocols such as peer-to-peer or client/server architecture
  - Media such as twisted pair wire, coaxial cables, or fibre optics

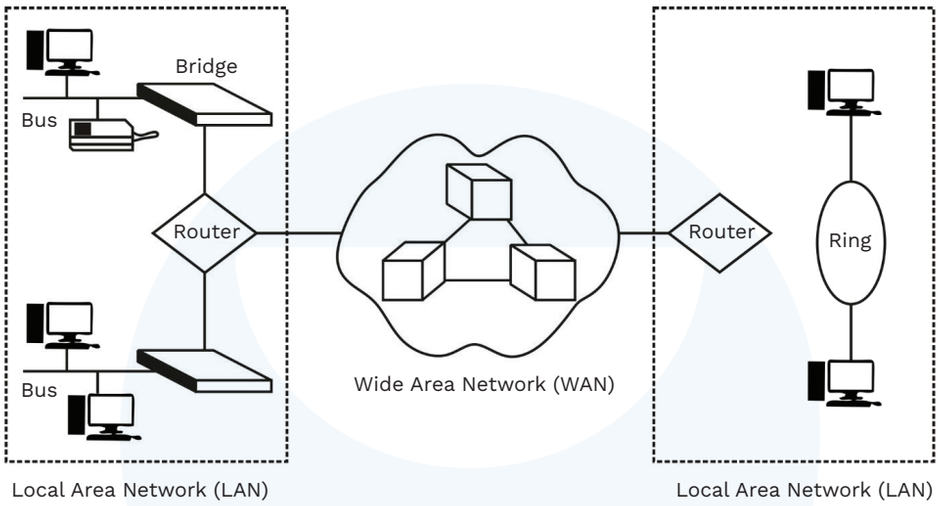


Fig. 3.7

Wide Area Network (WAN)

- A wide area network (WAN) covers a large geographical area as compared to LAN, spanning multiple states/countries. WANs are used primarily to connect dispersed sites.
- The main characteristics of WAN are
  - They have a diameter of the order of a few 1000 km.
  - Their bandwidth is very costly. Hence, bandwidth allocation plays an important role.

Metropolitan Area Network (MAN)

- A metropolitan area network (MAN) is a wide area network (WAN) that spans many cities. The underlying technology is similar to that of local area networks.
- High-speed communication between locations is made possible by MAN because it is based on a fibre-optic transmission technology. Both voice and data transmissions are possible.
- Data transmission speeds on a MAN may range from 34 Mbps up to 150 Mbps.

Personal Area Network (PAN)

- When many computers or other electronic devices (such as phones) are located within a short distance of one another (often less than 10 m), they may communicate with one another through a personal area network (PAN).
- Wireless or wired, PANs let devices share data and information or connect to a wider network, such as the internet.
- Whether or not the gadgets at issue are indeed owned by the individual in question is unknown. Typically, a PAN's range is only a few meters.

- When many electronic devices, such as laptops, mobile phones, and PDAs, are in close proximity to a single person, they are said to be in a PAN.

Internet

It stands for an international network of computers, i.e., it is that network that connects all the computers across the world. It is the world's largest WAN global network, the worldwide network of networks, backbones, and gateways, a window of the 'information superhighway' over the world. In today's world, the internet plays a huge role in every field. It is the fastest, easiest, and cheapest way for any user to get any information in the fields of business, education, or research in any field.

Hypertext Transfer Protocol (HTTP)

- For the World Wide Web (WWW) to function, data transmission must rely on HTTP. Application protocols for distributed, collaborative, hypermedia information systems are what the Hypertext Transfer Protocol (HTTP) is all about. It is a collection of non-linear items that creates a network via the use of hypertext connections.
- The HTTP is a standard for exchanging and transferring hypertext that may be used to enhance or facilitate connections among clients and servers thanks to the use of intermediary network components.
- Uniform resource identifiers (URIs) using either the HTTP or HTTPS URI scheme identify and find HTTP resources on the network. Hypertext Markup Language (HTML) documents with embedded URIs and hyperlinks create networks of connected hypertext resources.

Markup Language for Hypertext (HTML)

- Web pages and other information viewable in a web browser are often written in Hypertext Markup Language (HTML).
- Inside the text of a web page, HTML is represented by tags contained in angle brackets (such as <html>).
- Tags in HTML usually come in pairs, as </h1>, however, certain tags, called empty elements, appear alone. For every pair of tags, the first one is the beginning tag and the second one is the closing tag (they are also called opening tags and closing tags).
- Web designers may insert any kind of textual material (such as text, elements, remarks, and so on) between these tags. Elements of the HTML code are the fundamentals of every webpage. You can use HTML to create interactive forms that include media such as photos and objects.

Networking Developments

White-Fi

- White-Fi technology takes advantage of the unutilised space in the radio spectrum at TV broadcasting frequencies. Microsoft's experimental initiative to improve last mile internet access throughout the nation seems promising.
- It is common practise for television stations to transmit data using a bandwidth between 200 and 600 MHz. With better tropospheric propagation conditions, signals may be received from further away, but this can only happen if enough room is allowed between transmitters and receivers.
- Since these channels are underutilised in many regions, the spectrum is being used inefficiently. Nearly 93 per cent of India's available spectrum is now being wasted. To fill the gap in connection, White-Fi will utilise the available radio frequencies. Microsoft has assisted in bringing this technology to Kenya, Singapore, the United States, and London.
- The government's 'digital India' effort seeks to expand access to electronic services throughout the country by funding a nationwide fibre-optic network. Wireless technology may still be necessary to reach the end user, especially in remote locations, even if this network is established at the gram panchayat level.
- White Space refers to the unallocated portions of the radio spectrum, and it is being considered by several IT firms as a means of providing free last mile internet access to consumers.

Li-Fi

- There has not been anything quite like LiFi before, since the energy from the same source of illumination may be utilised for both lighting and data transmission. A subset of Optical Wireless Communications, LiFi (OWC). In addition to visible light, OWC also supports infrared and ultraviolet communications.
- Pure LiFi's Chief Science Officer, Professor Harald Haas, is credited with coining the name 'Li-Fi,' which stands for 'light based communications technology.' This technology uses light rather than radio waves to provide high-speed, bidirectional, networked, mobile communications.
- To transport data at gigabit rates, LiFi takes advantage of the electromagnetic spectrum's visible light region. This is in contrast to the use of conventional radio frequency (RF) transmissions used by Wi-Fi and other well established types of wireless communication.
- It is possible to send information by varying the brightness of a light source. After being picked up by a photosensitive detector, the data signal is converted from optical to electrical.
- Li-Fi may be used to offload data from Wi-Fi networks, but it can also be used to provide capacity for the larger downlink demand.

Project Loon

- Project Loon (c) is an R&D initiative led by Google X that employs a fleet of space bound balloons to bridge coverage gaps, restore service in disaster zones, and provide Internet access in underserved regions.
- Approximately 20 km above the Earth's surface are the project loon balloons. To go where they are required, loon balloons may either climb into the upper atmosphere or drop into the lower atmosphere and ride the wind currents in the appropriate direction.
- The balloons may be assembled into a massive communications network if the wind is blowing in the right direction. When the signal reaches the end of the balloon network, it is relayed to the global Internet below. They may use their phones or other LTE enabled devices to connect directly to the balloon network.
- In June of 2013, Loon innovation was tried in a trial program called Project Loon. The pilot test's findings, together with those of later tests conducted in New Zealand and Northeast Brazil, are informing the development of the project's future phases of technological development.
- The air pressure is just 1 per cent of that at sea level, and this gives less protection against ultraviolet radiation and severe temperature changes, which may drop as low as –80 degrees, making the stratosphere a challenging environment for engineers to work in.
- A balloon envelope, often constructed from polyethene plastic sheets, contains the inflating gas within the balloon. For a balloon to stay aloft for up to 100 days, a quality envelope is essential.

Internet of Things (IoT)

- The internet of things (IoT) is the inter-connection of uniquely identifiable embedded computing devices within the existing Internet infrastructure. Typically, loT is expected to offer advanced connectivity of devices, systems, and services that goes beyond machine to machine communications (M2M) and covers a variety of protocols, domains, and applications. loT is a scenario in which objects are provided with unique identifiers and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction.
- According to Gartner, there will be nearly 26 billion devices on the Internet of Things by 2020. ABI Research estimates that more than 30 billion devices will be wireless and connected to the Internet of Things (Internet of Everything) by 2020.

Cloud Computing

- A desktop computer cannot be used in isolation, as one has to go online for various reasons. Sharing of articles, notes, photos, or videos with colleagues, co-professionals, friends, relatives, etc., has been a long passion of net users, but online sharing has gone beyond this type of sharing. With the

increase in the speed and ease of availability of the internet, online remote servers have become a new medium to store data and retrieve the same just like a hard disk of a desktop computer.

- Cloud computing is a technology that uses the internet and central remote servers to maintain data and applications. It allows consumers, businesses, educational institutions, etc., to use applications without installing costly software on their computers. For example, to send an email, all one needs is just an internet connection. The server and email management software are all on the cloud and are managed by cloud service providers such as Yahoo, Google, etc.
- Here is where cloud computing comes in handy: rather than setting up a whole suite of software on each machine, users need to simply set up a single application. This programme would enable all workers to access a central server where all the necessary software for their jobs is stored. Computing in the cloud Changes in workload are substantial in a cloud computing environment.
- To put it another way, local computers are no longer required to perform all the work when it comes to operating programs. Instead of humans managing them, the cloud's distributed group of computers takes care of everything.
- However, consumer input on hardware and software requirements is reduced. The user's machine does not need to be capable of much beyond running the cloud computing system's interface software, which may be as basic as a web browser.
- Email services that require users to access their accounts through a web browser, such as Gmail, Yahoo, etc., are also instances of cloud computing. In this case, the user's PC is not required since all of the necessary software and data storage are hosted by the service itself in the cloud.

Assumptions and Design Considerations for a Cloud Based System

- If you want to understand the inner workings of a cloud computing environment, it is best to break it down into two parts: the front and the back. The 'front' of a computer is the side that faces the user. The 'cloud' component of the system serves as the back end. The user's computer and the software needed to access the cloud computing system make up the front end.
- Not all cloud computing solutions have the same user interface. Web based email services, for instance, make use of pre-existing web browsers such as Internet Explorer and Firefox.
- Other systems offer unique applications that enable network connectivity to customers. Back end components include the network of interconnected computers, servers, and data storage systems known as the 'cloud.'
- From data analysis to entertainment, in principle, any computer software may be part of a cloud computing system. A separate server is often used for each program.
- Every aspect of the system is managed by a single server that keeps an eye on traffic and user needs. Protocols are the regulations it abides by, while middleware is the specialised software it employs. Using middleware, machines in a network may exchange information with one another.
- The majority of the time, server resources are underutilised. This implies that plenty of computing resources are being put to waste. A single physical server may be tricked into believing it is part of a cluster of servers by installing numerous operating systems on it. The term 'server virtualisation' describes this method. By increasing each server's productivity, server visualisation helps cut down on the requirement for additional hardware.
- To meet the storage needs of their customers, some businesses may need hundreds of digital storage devices, which is why a cloud computing provider with a broad client base is desirable.
- To safely store all of its users' data, cloud computing systems need twice as many hard drives as they now use. To protect client privacy, a cloud computing system must duplicate data and keep several copies in many locations.

- To protect client privacy, a cloud computing system must duplicate data and keep several copies in many locations.
- By making duplicates, the primary server may get access to the backups and recover information that would otherwise be inaccessible. Redundancy refers to the practise of creating extra copies of data for safety purposes.

Crucial Aspects of Cloud Computing

- Accessible through a wide variety of network protocols, 'cloud capabilities' may be used with a variety of thin and thick client devices.
- Email, apps, and server services are all examples of the kinds of computer services that may be given on demand without any involvement from a human service provider. Cloud providers such as Microsoft, Google, and IBM all provide self-service options on demand.
- Using a multiple tenant approach, the supplier pools its computing resources to serve several customers, allocating and reallocating various types of hardware and software on an as needed basis.
- Among the resources available are data storage, computing power, memory, network throughput, virtual machines, and electronic mail. When resources are combined, economies of scale may be achieved.
- When using cloud computing, both the service provider and the person using the service may get accurate information on how much of the service has been used.
- In the cloud, services may be supplied fast and elastically, sometimes automatically, to scale out and released just as quickly to scale in. Most consumers have the impression that they can buy as many of the offered capabilities as they need, whenever they want.
- Cloud computing services employ a metering function that helps to regulate and optimise resource utilisation. This means that IT services are paid for on a 'pay as you go' basis, similar to how utilities such as electricity and municipal water are priced. Costs will increase proportionally with use.
- That cloud services may expand or contract in response to fluctuating demand is known as 'scalability.' Since cloud computing makes it simple to scale up the number of users accessing a given service on a given server, this is a significant benefit of the cloud.
- Segmentation, governance, policy driven enforcement, isolation, and levels of service for diverse consumer groups are all examples of 'multitenacity,' or the requirement for individualised service tailored to each customer's specific needs. Consumers may use a cloud infrastructure company's service offering or may be from the same company but operate as autonomous divisions inside the larger whole.

Cloud Services

- Users interact with cloud computing environments through the services that the cloud computing environment provides. The following examples are of services that are provided by a cloud (cloud services).
  - Virtual servers
  - Database services
  - Email applications
  - Storage
- A business may either employ public cloud services offered by a third party, or it can create its own private cloud. To this end, the corporation may make its cloud based services available to employees, certain clients and partners, and the general public. The following criteria should be present in any service that claims to be a 'cloud service.'
  - To be available online or via a company's internal network
  - Allow users to handle their provisioning
  - Allows for quick and easy resource flexibility in response to fluctuating user demand
  - Delivers resources from a pool without requiring the user to be aware of its existence
  - Allows for metering to take place, which makes a flexible chargeback scheme possible
  - The ability to examine the health of cloud resources in a graphical dashboard is provided in point

- The infrastructure that supports cloud services makes use of two essential enablers to deliver these features.
  - 'Automation' enables elastic utilisation of available resources and the migration of workloads to where they are available. It also allows for the supply and removal of service instances, which helps scale.
  - The ability to pool and dynamically distribute computer resources is the primary benefit of 'virtualisation.' Additionally, it paves the way for the introduction of pay-as-you-go pricing models.

Cloud Service Models

Service models are sometimes referred to as 'delivery models' because they describe the services that are delivered by the cloud model.

Software as a Service (Saas)

- In the SaaS model, a software provider licences a software application to be used and purchased on demand. This service runs on the cloud and multiple end users use it. It runs on a web browser, for example, Gmail, a popular SaaS product. It is usually billed based on usage and has a multi tenant environment.
- Applications can be accessed through the network from various clients (web browser, mobile, phone, etc.) by application use. It does not require client installation but just a browser or other client device and network connectivity. Traditional application modification is supported by SaaS apps in the same way. The configuration choices are not locked in and may be changed by individual customers, unlike more conventional business software. There is room for customisation in the available configuration choices, and each client may have their own preferences.
- In many SaaS services, the client may choose to have their logo and, in certain cases, a colour scheme implemented via the user interface or by working with the application provider's team. Using a preset of configuration choices, the programme may be tailored to the exact specifications of the user. For instance, this might be done to accommodate the widespread desire amongst customers to have their own brand's aesthetic reflected in their chosen applications.
- Characteristics of SaaS: it is important for cloud computing to ensure that solutions sold as SaaS comply with generally accepted definitions. Characteristics of SaaS include:
  - Commercial software for web access
  - A central location, managed by software
  - 'One to many' model of software delivery
  - To handle software upgrades and patches, users are not required
  - APIs allow integration between different pieces of software
- SaaS's pay-as-you-go price structure makes it accessible to start-ups and sole proprietors.
- One of the main benefits of using a SaaS service is how simple it is to access and utilise.
- Data loss is reduced in applications due to cloud storage.
- SaaS applications have the advantage of being quickly scalable to accommodate fluctuating user demand. Customers may scale up without worrying about investing in new server hardware.
- By having everything in one place, teams can work together more effectively.
- Compared to their more cumbersome predecessors, conventional software, SaaS apps have several advantages. In other words, there is no need for end users to deal with binary code installations or removals. Because SaaS apps are distributed via the internet, they may be used on a broad range of platforms.
- One of the most notable features of modern software is how often and dynamically it can be updated.
- The agile methodology life cycle is favoured by the SaaS model.
- In terms of velocity, changes occur far more quickly in SaaS applications.

Platform as a Service (PaaS)

- The virtualised servers and related services are rented out to the user so that they may run their current software or create, test, and deploy new software using the service delivery model. Hardware, software, data

storage, and data networks may all be rented through a 'platform as a service' (PaaS).

- The characteristics of PaaS are
  - Customisable/programmable user interface
  - Unlimited database customisations
  - Robust workflow engine/capabilities
  - Granular control over security/sharing (permissions model)
  - Flexible 'services enabled' integration model
- With certain PaaS solutions, even 'non experts' can create their own custom software. All it takes is a single click on their computer browser.
- The ability to 'rent' virtual infrastructure provides financial and operational advantages over investing in physical infrastructure. They may avoid spending money on hardware or hiring specialists to run it. This frees up their time, allowing them to put more effort into creating new apps. In addition, customers will be able to save money by renting the resources they use rather than purchasing permanently installed but underutilised capacity.
- To adapt to new conditions, features might be altered. Because of this platform's adaptability, users are free to customise it to meet their unique needs and decide for themselves which features will be included. Customers may 'cherry pick' the functionalities they deem essential.
- Data backup and recovery, as well as general security, are supplied as part of the service.
- Distributed teams can collaborate effectively, since a single application build may be worked on by developers from all over the world as long as they have access to the internet and a web browser.

Infrastructure as a Service (laaS)

- Infrastructure as a service (IaaS) is a model in which an organisation outsources the equipment used to support operations, including the storage, hardware, servers, and networking components. The service provider owns the equipment and is responsible for housing, running, and maintaining it. The client typically pays on a peruse basis. laaS is sometimes referred to as hardware as a service (HaaS).
- Automation of administrative tasks, dynamic scaling, desktop virtualisation, and Internet connectivity are the various characteristics or we can say components of an Infrastructure as a Service. The figure shows the various service models of cloud computing and their interrelations.

Cloud Delivery Models

- Where the data resides and who manages the cloud service are all factors that may be classified under 'cloud delivery models,' which describe how a given business implements and makes use of a cloud computing service. The capabilities required by a cloud solution may be delivered through any of the many delivery modes supported by cloud computing.
- Here are some examples of how cloud services are delivered: the four types of cloud are private, public, community, and hybrid.

Private Clouds

- An example of a cloud solution, 'private clouds' are those in which the underlying hardware and software are set up specifically for the needs of a single company. An organisation often functions as a cloud service provider to business units, allowing them to reap the advantages of the cloud without the overhead of provisioning their infrastructure. Consolidating and centralising services in the cloud allows the company to take advantage of economies of scale and centralised service administration.
- Many businesses choose not to use the public cloud because of the benefits that a private cloud may provide. The cloud's resources become more under the company's direct control. When tasks cannot be completed efficiently in a public cloud due to factors such as network latency, regulation, or security concerns, private clouds are the way to go.
- Organisations may own, manage, and operate their own private clouds, or hire a third party to do so. The private cloud infrastructure is often set up inside the company's facilities.

However, it may alternatively be housed in a third party data centre.

Public Clouds

When the cloud computing platform is made accessible through the Internet to the general public or a large industry group, we refer to this model as a 'public cloud.' The user does not control the underlying infrastructure; rather, it is held by the company offering the user access to the cloud. Some service models include freemium, subscription, and à la carte pricing.

Community Clouds

- The cloud architecture for a community cloud is shared by several organisations to better serve the needs of that community.
- The major objective of a community cloud is to allow participating enterprises to enjoy the cost savings and flexibility of the public cloud while still enjoying the security, privacy, and compliance with policies often reserved for private clouds. Participating businesses or a third party may offer and operate the community cloud infrastructure, which may be hosted either on premises or at a third party's data centre.

Hybrid Clouds

- Despite each hybrid cloud's individuality, they are tied together by data and application mobility technologies.
- Using a hybrid setup, businesses may reap the benefits of public cloud computing without having to open up sensitive data and apps to the public. Payment processing for customers is a private cloud service, and payroll for employees is a secondary cloud service that may run well in a well designed hybrid cloud.
- The difficulty of efficiently constructing and regulating such a system presents a problem for hybrid clouds. Implementation is challenging because of interactions between private and public components and the need to receive and supply services from several sources as though they came from a single source.

Severs in the Cloud

- To put it simply, servers are what makes cloud computing possible. One method through which cloud service providers make available servers to their customers is platform virtualisation.
- Platform virtualisation is a technique that allows numerous operating systems to be loaded on a single server by partitioning the server's actual hardware resources into many virtual computing environments.
- The virtual machine's hypervisor is its brain. The software mediates between hardware and software. System resource allocation is its primary function. Each of these Operating Systems is running on its dedicated server. To be precise, there are three distinct virtual machine images underlying these three OSes. Each machine image is a time-stamped record of a certain operating system, and once put into the virtual machine environment, functionality is restored to the original OS.
- You should know that your payment is not for a full server. Instead, cloud service providers provide you access to a virtual computer image, which is only one of many that are stored in the cloud's virtualised server farm.
- Cost-effective, fast deployable, and scalable servers may be provided by cloud computing service providers because of the platform virtualisation capabilities that make this possible.
- These characteristics include, in particular, in cloud computing, hardware incompatibilities are no longer a concern for service providers or their customers, thanks to the elimination of hardware requirements. Just make sure they have laser-like concentration on their software and SLAs. Moreover, developers have nothing to do other than consider which operating system would best serve their needs and how much power they will need.
- Without platform virtualisation, the average server in a corporate data centre only runs at a utilisation rate of 5–10 per cent. Even at their peak, most systems can only reach around 20 per cent capacity. Cloud service providers take advantage of this by using

platform virtualisation to host several virtual servers on a single physical server, allowing them to lease out more servers (virtual) while using the same amount of hardware.

- Virtualisation makes it simple to install a 'server,' since all one has to do is load the virtual machine image. A picture of a machine may likewise be easily shattered.
- Software dependencies are eliminated since the virtual machine environment takes care of things like installing required device drivers.

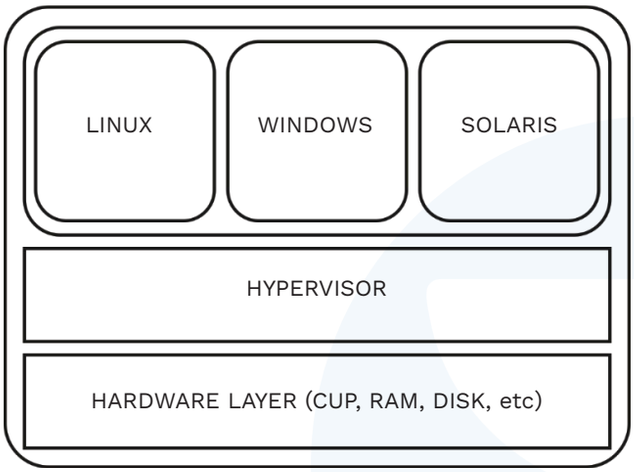


Fig. 3.8

Grid Computing

- By using a distributed network of computers, or 'grid,' this new paradigm of computing allows for increased computational throughput. A concept of virtual machine architecture is built that can execute a task in parallel.
- To address massive computing problems, grids pool the processing power of several computers through a network (often the Internet). To simulate a parallel division of labour amongst processes, grids allow for many more calculations to be performed at once than would be feasible on a single computer, or allow for the processing of enormous data sets by tearing them down into several small pieces.
- Unlike conventional computer clusters, it may facilitate computing across different administrative domains.
- One of its main features is a methodology for tackling large-scale computing problems by pooling the idle processing power and data storage capacity of many individual computers located in different locations and connected through a network.
- Sharing heterogeneous resources (those built on various platforms, hardware/software architectures, and computer languages) over a network utilising open standards is a key component of this model. For the time being, we only know that it has something to do with virtualising computer resources.
- The fourth reason is that it can handle numerous smaller issues at once while still being able to solve problems that would tax even the most powerful supercomputer. As a result, it supports several users simultaneously.
- Grid computing requires a complex ecosystem of low-cost and high volume processors; 'blade server' technology; storage options such as storage area networks; a cost-effective and enterprise ready operating system; and connectivity technologies such as ethernet or fibre channel, all of which are currently beyond the capabilities of any single player.
- Version 109 of Oracle's flagship database is touted as the first relational database designed specifically for grid computing. Snapshot, developed by storage networking pioneer Network Appliance (Net APP), enables several copies of data to coexist inside a single 'physical data block.'

Applications of Grid Computing

- By using distributed computing resources, issues of 'grand challenge' scope, such as protein folding and financial modelling, may be tackled. Organisational earthquake and climate modelling using IT resource optimisation.
- Small and big businesses alike now understand that to stay competitive, they must invest in better computing and storage infrastructure as well as streamline their operations. Here are a few possible uses.
    - Because of their limited budgets, educational institutions should consider the more cost-effective alternative of a cluster of conventional computers, which can attain teraflop performance via judicious allocation of the work at hand.

- Meteorology and climate modelling provide a technique to address enormous computational challenges by pooling the idle processing resources of thousands of computers throughout the globe.
- Agencies in the public sector may utilise it to pool together dispersed high-performance computing capabilities to take on national projects such as updating voter lists and tax databases.

The Indian Contingency

- Indian businesses are ahead of the curve when it comes to anticipating a transition to a 'Grid IT environment,' with over half of those surveyed in an Oracle study saying they had either already made the decision to use grid technology or were in the process of doing so.
- Based on the findings of the most recent Oracle Grid Index study, it seems that the importance of enabling technologies is now being recognised. The Grid Index is a measurement of how well an organisation is prepared for, interested in, and has adopted grid computing. India Oracle Grid Index increasing from 2.9 to 4.4, Oracle's Grid Index shows the greatest rise of any grid index in the world, which is close to 52 per cent.

Transmission Modes in Network

Interfaces are bundles of wires used to connect devices. If two or more digital devices are connected, then data transmission occurs when the bits transfer from one device to the other through physical media. It also specifies if bidirectional data transmission is possible or whether devices must take turns transmitting and receiving data. One may transmit data in either a serial or parallel fashion.

Transmission in Parallel

- By utilising many lines at once, the parallel transmission allows for the simultaneous transmission of multiple bits. It is normal to practise for data to be sent between two devices in parallel when the distance between them is small.
- Advantages of parallel transmission:
  - To talk across a short distance.
  - Second, the rate at which information may be sent is increased.
  - Third, packets of data are sent all at once.
- Disadvantages of parallel transmission:
  - One of the drawbacks of parallel transmission is that it is more expensive.
  - Second, many channels of communication are needed.
  - Third, to prevent signal deterioration while sending data over greater distances, heavier gauge cables must be used.

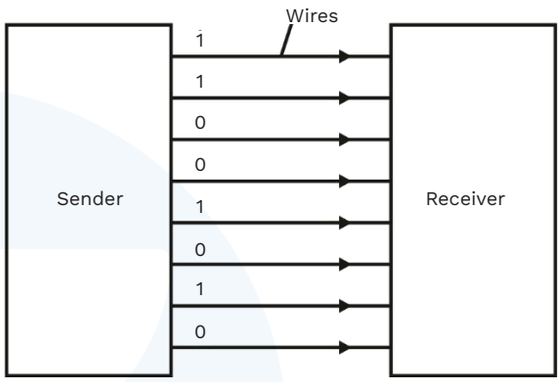


Fig. 3.9

Transmission in a Serial Fashion

- The bits are sent down the single line one at a time in serial transmission. transmission of the least significant bit first. This is because serial transmission is both more cost-effective and dependable. Because individual bits are delivered, data transfer rates are poor.
- Synchronous and asynchronous methods of providing serial communication exist. There is an extra layer of complication between the transmitter and receiver.
- Advantages of serial transmission:
  - Serial transmission is well suited for long distance communication and so offers several benefits.
  - It is more affordable for one thing.
  - It can only function with a single line of communication.
  - Finally, it is more consistent.
- Disadvantages of serial transmission:
  - Its performance varies depending on the data rate.

- It takes a long time for information to be sent.

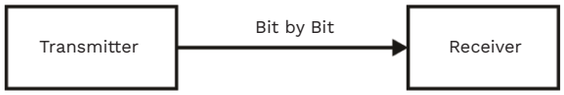


Fig. 3.10

World Wide Web

When it comes to publishing and reading hypertext content online, the World Wide Web or web is the software application of choice. When several computers are linked together in a network, we call it the Internet.

- The data on the internet travels through a computer network. As a result, the Web relies on the global interconnected computer network that is the Internet and the HTTP protocol to function. Web browsers are multiprotocol, meaning they can communicate with many Internet resources, and the number of protocols supported by browsers is growing constantly. This is the main distinction between the World Wide Web and web browsers.
- Accessing linked content on the World Wide Web (WWW) is similar to navigating a massively distributed system with millions of clients and servers. Document collections are stored and managed by servers, while clients provide users with a simple interface to view and interact with those documents.
- The globe wide web is a massive client server system comprised of millions of servers located in different parts of the world. Each server has its own document repository, with files for each item in the collection.
- When a client requests a file, the server retrieves it and sends it on to the user. New document storage requests may also be taken in.
- Uniform resource locator (URL) references make it easy to point to specific pages inside a document. The document's location is specified, usually by including the server's domain name system (DNS) name and the file name by which the server can find the document in its local file system. As a bonus, a URL may also be used to denote the application level protocol that will be used to transmit the file across the network.
- A client uses a specialised programme called a browser to communicate with Web servers. A browser's job is to render a page accurately. A browser also lets the user pick a link to another document, which the browser then retrieves and displays. The diagrammatic representation of the whole system.

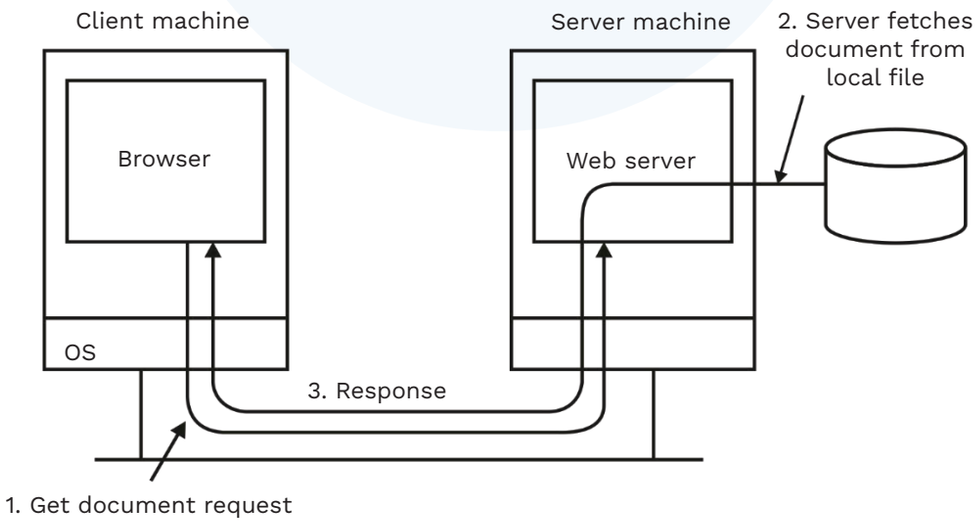


Fig. 3.11

Web Services Architectures

- To put it simply, a Web service system consists of a service producer (provider), a service consumer (requester), and a Web service registry. There is a connection between the provider, who offers the service's interface and implementation, and the requester, who employs the Web service.
- The three primary responsibilities of a web service are
  - To begin with, let's define what we mean by 'web service provider' (WSP). It is the service provider's responsibility to put the service into action and make it accessible over the web.
  - To make use of a web service, a client simply establishes a data connection and issues an XML request.
  - When creating a new service or searching for an existing one, developers may use the register as a one stop shop. It is like a hub where all the different businesses and services may be coordinated with one another.

Web Modelling

- There are several possible forms of written communication. In contrast to the barebones ASCII text files that make up some documents, others are conveyed using a set of scripts that are run in response to the page being downloaded into a browser. Most importantly, though, is that one document may refer to another.
- Hyperlinks are used for this purpose. Hyperlinks to other documents may be made more obvious to the user when the page is shown in a browser. To choose a link, the user just clicks on it. When you click on a link, your browser sends a request to the server where the linked file is kept.
- It is then sent to the user's computer, where it is received by the browser and displayed. There are two possible outcomes: the new document may either replace the existing one or open in a pop-up window.
- Most Web pages are represented using a particular language called HTML. HTML is a markup language, which means it has built-in keywords for making divisions in a document. An HTML the document, for instance, will have a header and body. Tables, headers, forms, and lists, all have their distinct markup in HTML.
- Images and animations may be included in a document at certain points as well. In addition to these building blocks, HTML also contains a wide range of keywords that tell the browser how to render the content. You may use keywords to format your text in many ways, such as changing the font, changing the size of the text, making it bold or italic, aligning it, and so on.

Web 1.0

- The World Wide Web (WWW) is a collection of hypertext documents that may be viewed online. In web 1.0, the content was produced by a small group of authors for a huge audience. Therefore, anyone may get information by going straight to the source.
- There was hardly any participation or content creation from users. Web 1.0 is the first version of the World Wide Web, and it was a 'read-only web,' as Berners-Lee put it. That is to say, we could use the early web to do searches and consume content. On the other hand, this is precisely what most website owners were hoping to achieve: to have their material readily accessible to anybody, at any time.

Web 2.0

- The term 'Web 2.0' refers to an enhanced version of the World Wide Web. Weblogs (blogs), social bookmarking, wikis, podcasts, Rich Site Summary (RSS) feeds (and other kinds of many to many publishing), social software, web application programming interfaces (APIs), and online web services such as eBay and Gmail all provide advantages over static, read-only websites.
- The term 'Web 2.0' is more of 'a notion in people's brains than a reality,' as Stephen Fry puts it. In this scenario, the giver and taker relationships are emphasised. As a result of being able to both upload and download, there is 'true interaction,' if you will.
- Although the name indicates a technological upgrade, Web 2.0 describes a shift in how

Web content is created and consumed by both programmers and regular people. Web 2.0 is an umbrella phrase for several changes to the Web and a generalised shift in user behaviour. We might think of this change as a progression from just consuming material on the Internet to actively contributing to, and participating in, the process.

- Web 2.0 is a term that defines the evolving trends in the usage of World Wide Web technology and Web design that strives to stimulate creativity, secure information sharing, encourage collaboration, and improve the functioning of Web 1.0. These have led to the establishment and expansion of social networking websites and video sharing websites such as Facebook and YouTube, etc.
- Sites built using Web 2.0 technologies enable visitors to do more than merely research a topic. To execute software applications, users may now depend on 'network as platform' computing, which is built on the interactive features of Web 1.0. Users of a Web 2.0 site may both contribute to and edit the site's content. These websites have what is called an 'architecture of participation,' which allows users to actively contribute to the application. The content of conventional websites can only be changed by the site owners, and users may only see it in a passive capacity. Web 2.0 Websites often incorporate some of the following features/techniques.
    - The links provided are pointers to relevant resources. The most popular and high quality pages get the most links.
    - To get what you need quickly and easily, search engines like Google have made great strides in recent years.
    - Tags allow users to organise material for searching and discovery without being constrained by predetermined categories.
    - Content may be co created by several people and updated in real-time, thanks to authoring. In wikis, users may undo and redo previous edits, creating material that is both dynamic and iterative. Blogs are cumulative because the contributions of many different people are added up over time.
    - When the material is updated, subscribers are alerted using a model built on the RSS (Really Simple Syndication) protocol.
    - Alternately, algorithmically automated pattern matching for individualised service i.e., Amazon.com recommendations.

Web 3.0

- The phrase 'Web 3.0' is used to describe the many ways in which the way we use and interact with the Internet has progressed.
- These include the use of AI technology (the Semantic web, the Geospatial web, or the 3D web), the web being converted into a database, content being made available by many non browser apps, and so on.
- The term 'Web 3.0' refers to the transformation of the World Wide Web into a database as part of the next stage of the Web's development and user interaction. After 10 years of focusing on the front end of the Internet, we have entered the period of Web 3.0, in which we will modernise the back end. One definition of Web 3.0 is the 'read-write-execute' web.
- Web 3.0 may be thought of as the next generation of the Internet, where talented people use web 2.0 tools to create innovative new forms of content and services.
- The terms 'website' or 'webpage' becomes obsolete in Web 3.0, data are shared rather than owned, and several perspectives on 'the same web/the same data' are presented by various services.
- These services, which may take the form of software (web browsers, MMOs, or anything else), hardware, or anything else, must prioritise context and personalisation and will do so via the use of vertical search.

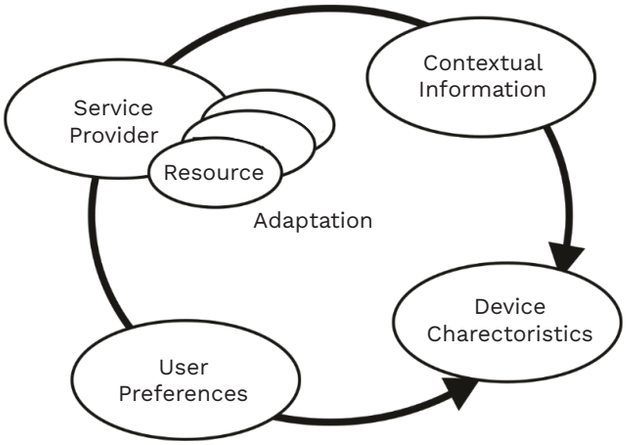


Fig. 3.12

Web Service

- It is a service offered by an electronic device to another electronic device, communicating with each other via the world wide web.
- In a web service, Web technology such as HTTP, originally designed for human to machine communication, is utilised for the machine to machine communication for transferring machine-readable file formats such as XML and JSON.
- A web service is a client-server application for communication. It is a method of communication between two devices over a network.
- It is also a collection of standards or protocols for exchanging information between two devices or applications. For example, JAVA applications can interact with JAVA, Net, and PHP applications. So, web service is a language independent way of communication.

Simple Object Access Protocol (SOAP)

- Pronounced 'simple object access protocol,' this is a method of exchanging data between devices. It is a three part protocol written in XML.
  - First, there is the envelope, which specifies the message's format and handling.
  - A standard way of depicting the invocation and completion of a set of predefined procedures.
  - Instances of application-defined datatypes are encoded according to a set of criteria.
- There are three distinguishing features of the soap.
  - Since SOAP is protocol independent, it may be used with any transport mechanism, including the widely used HTTP, SMTP, TGT, UDP, and JMS.
  - The ability to grow: security and WS-routing are among the extensions under development.
  - SOAP is not tied to any certain programming paradigm.

Rich Site Summary (RSS)

RSS: It enables publishers to syndicate data automatically. It benefits users who want to receive timely updates from their favourite websites or aggregate data from many sites. It removes the need for the user to manually check the website for new content. Their browser constantly monitors the site and informs the user of any updates.

Extensible Markup Language (XML)

Extensible markup language. It defines a set of rules for encoding documents in a format that is both human-readable and machine readable. The design goals of XML emphasise simplicity and usability across the internet. It is a textual data format with strong support via Unicode for different human languages.

Advantages of Web Services

- Service Reuse extends the concept of code reuse. Each domain specific function is written just once and then used by many different applications.
- Loose coupling means that the application's services may function apart from one another. The application may be adjusted independently of one another.
- Creating 'silos' of data where it cannot be accessed by other programs is a major hindrance to integration. Web services provide the connecting tissue that facilitates more effective internal and external communication.

Network Security

- Network security refers to activities designed to protect the usability and integrity of networks and data. It targets a variety of threats and stops them from entering or spreading on the network.
- Working: it combines multiple layers of defences at the edge and in the network. Each network security layer implements policies and control. Authorised users gain access to network resources but malicious actors are blocked from carrying out exploits and threats.

Types of Network Security

- Malware, which includes viruses, worms, frozen files, ransomware, and spyware, may be thwarted with the use of antivirus and antimalware programs. Malware may infiltrate a network and then sit dormant for days or weeks. In addition to an initial check for malware upon launch, a top-notch anti-malware tool should persistently monitor files for the presence of malware, then either delete the offending code or repair any harm it may have caused.
- When used properly, NAC (Network Access Control) may effectively quarantine or severely restrict access to the internet for any non compliant endpoint device.
- According to the field of Behavioural Analytics, 'abnormal behaviour' is defined as any pattern of action that deviates from the norm. Behavioural analytics is a method for identifying actions that are not typical.
- Application security: any program or application might have security flaws that would allow hackers access to the system. Protecting an application often entails using a combination of hardware, software, and procedures designed to plug these holes.
- Email security: when it comes to security vulnerabilities, emails provide the greatest risk. Phishing campaigns are created by hackers using social engineering techniques to get users' sensitive information and then direct them to malicious websites. An email security program may prevent the disclosure of private information by intercepting incoming assaults and monitoring outgoing mail.
- DLP: with DLP, or data loss protection, you can be certain that private data will never leave your local area network. Avoids the risk of sensitive data being uploaded, sent, or printed by untrusted sources.
- Intrusion prevention systems (IPS) are designed to actively prevent attacks by monitoring network traffic.
- Firewalls: fire barriers prevent unauthorised users from accessing a company's trusted internal network.
- Extranet: a network that is not part of your local area network, such as the internet. To accept or deny traffic, they follow a predetermined set of guidelines. Depending on the needs of the organisation, a firewall may be hardware based, software based, or even both.
- It is a security solution for private networks that detects and stops intrusion attempts.
- Its implementation is not limited to either hardware or software alone. To stop hackers and other malicious users from accessing private networks through the internet, firewalls are set up between the network and the internet.
- The firewall monitors all incoming and outgoing data packets across the network and discards those that do not pass predetermined security muster.
- VPN**:** By using a virtual private network (VPN), data travelling between a user's device and a remote network may be protected while travelling across an insecure public network like the internet. To verify the identity of the sending and receiving devices, a remote access VPN makes use of SSL encryption.
- Mobile device security: cybercriminals are increasingly focusing on mobile devices and applications; thus, it is important to implement mobile device security measures to ensure that all network traffic remains private.

Beebone Virus

- Belongs to Trojan malware, which gets privileged access to a user's computer by taking its identity and deploying smart and corrupt techniques to attack vulnerable computers.

- It intrudes into windows silently and seventy. It changes the same desktop icons and systems registry. Beebone virus constantly changes the names of its files so that this virus can escape from the removal of antivirus programs.
- The two windows virus known as vobfus and beebone, are regularly found together.
- Vobfus is first to first to arrive on a machine and then uses different tactics to infect victims.

Botnet Spam

Travels through internet connections: It is a combination of two words robot and network. These two words imply how a Botnet travels through an internet connection.

Risks due to Botnet Spam

- Criminals use the botnet to send out spam email messages and spread viruses, attacking computers and servers while committing other types of crimes and grand.
- If a computer becomes a part of a botnet, the concerned computer might inadvertently end up helping criminals.
- Botnets are exploited for numerous services such as mixing bitcoins, click fraud, theft of application serial numbers, logins, and financial data such as credit cards, debit cards, etc.

Practice Exercise

1. With reference to 'Li-Fi,' recently in the news, which of the following statements is/ are correct?

I. It uses light as the medium for high-speed data transmission.
II. It is a wireless technology and is several times faster than 'WiFi.'

Select the correct answer using the code given below.

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

2. 'Project loon,' sometimes seen in the news, is related to:

(A) Waste management technology
(B) Wireless communication technology
(C) Solar power production technology
(D) Water conservation technology

3. What will be the possible impact of project loon?

I. It will create a monopoly for google as an internet provider.
II. It will provide internet access to rural and remote areas.
III. It will affect the weather system.

(A) I and II
(B) Only II
(C) I, II, and III
(D) I and III only

4. 'Internet of things' sometimes seen in the news is related to:

(A) A transfer of data over a network without requiring human to human or human to computer interaction.
(B) Buying of products with the help of the internet.
(C) Online monitoring of agricultural markets.
(D) Create things and products with the help of 3D printing and the internet.

5. MAC stands for:

(A) Message authentication code
(B) Message authentication connection
(C) Message authentication control
(D) Message authentication cypher

6. Web mail, web analytics, e-hospital, and biometric attendance system are mainly examples of which amongst the following types of cloud computing?

(A) Infrastructure as a Service (laaS)
(B) Platform as a Service (PaaS)
(C) Software as a Service (SaaS)
(D) None of the above

7. Which amongst the following statements is/are correct about the cloud computing models based on accessibility?

I. Public cloud is best suitable for taking information and communication technology to the masses.
II. Private cloud is the most secure as well as least expansive.
III. Community cloud focuses on the need of the user group.

Select the correct code.

(A) I and II only
(B) II and III only
(C) I and III only
(D) I, II, and III

8. Which amongst the following can work even without the availability of internet services?

(A) Cloud computing
(B) Wide area network
(C) Local area network
(D) None of the above

9. Consider the following statements regarding cloud computing.

I. Private cloud describes a 'cloud services that can be accessed only by a pre defined group of people.
II. Private cloud provides elastic behaviour as well as scalability.
III. Scalability is that feature of cloud computing that alters the service to change in size or volume to meet the user's needs.
IV. A cloud environment can be accessed from anywhere in the world so long as the user has access to the internet.

Which of the above statements are correct?

(A) I, II, and III only
(B) I, III, and IV only
(C) II, III, and IV only
(D) I, II, III, and IV

10. Which of the following is a software distribution model in which the applications are hosted by a service provider and are made available to users over a network?

(A) Information as a service
(B) Platform as a service
(C) Software as a service
(D) Platform and software jointly as a service

11. Consider the following statements about project LOON in India.

I. It will provide connectivity in rural and remote areas.
II. It will place a network of balloons in the stratosphere.
III. Solar panels will be used to power electronic equipment in the balloon.

Which of the following statements is/are true?

(A) I and II only
(B) I, II, and III
(C) II and III only
(D) I and III only

12. Assertion (A): Facebook's free basics were violating the net neutrality principle.

Reason (R): net neutrality is nothing but the operator and the government should treat all data on the internet unequally.

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.
(E) Both A and R are false.

13. 'Frequency hopping' is more useful for military communication due to which amongst the following main reason?

(A) Quick connectivity with multiple devices comes at once.
(B) It is more cost-effective.
(C) Such communication is difficult to detect.
(D) No above reason correctly explains it.

14. Which amongst the following statements is correct?

(A) Signal-to-noise ratio is poorer in fibre optical cables as compared to copper wires using electron based data transmission.
(B) Power transmission lines have a much better signal-to-noise ratio as compared to microwaves.
(C) Li-Fi is much faster than Wi-Fi.
(D) None of the above statements is correct.

15. As head of some big engineering project if you are entrusted with the responsibility of taking internet services to rural areas, then which amongst the following may be viable and feasible usages?

I. Using balloons like in project loon
II. Use of very small aperture terminals (VSATSs)
III. Use of fibre optic cables

Select the correct code.

(A) III only
(B) I and III only
(C) I and II only
(D) I, II, and III

16. Firewalls operate by:

(A) The pre-purchase phase
(B) Isolating intranet from extranet
(C) Screening packets to/from the network and providing controllable filtering of network traffic
(D) None of the above

17. An intranet is:

(A) A network operating system
(B) A network filing system
(C) A private company network
(D) A private company network that includes external specified users

18. Assertion (A): the Internet of things (IoT) helps develop smart manufacturing.

Reason (R): the IoT is device to device communication without requiring human intervention.

(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true, but R is not the correct explanation of A.
(C) A is true, but R is false.
(D) A is false, but R is true.
(E) Both A and R are false.

19. Consider the following statements regarding cloud computing.

I. Hybrid cloud takes benefits of the private cloud by having better control over the resource and the public cloud by having more scalability.
II. Broad network access refers to access to cloud service by various platforms such as mobile phones, laptops, and PDAs.
III. Cloud computing suffers from legal provisions procedures and data security.

Select the correct codes.

(A) I and II
(B) I and III
(C) I, II, and III
(D) None of these

20. Which of the following statements is/are correct about Li-Fi?

I. It can transmit data at the speed of light.
II. Over a period of time, Li-Fi technology could replace Wi-Fi.

(A) I only
(B) II only
(C) Both of the above
(D) None of the above

21. Consider the following statements about Project Loon in India.

I. Project loon will use LTE technology to provide connectivity in rural and remote areas.
II. It will place a network of balloons in the stratosphere.
III. Solar panels will be used to power electronic equipment in the balloon.

Which of the above statements is/are correct?

(A) I and II
(B) II and III
(C) I and III
(D) I, II, and III

22. Firewalls are used on PC for:

(A) Security
(B) Authenticity
(C) Data transmission
(D) Data division

23. Half duplex system for communication has:

(A) Communication in a single direction.
(B) Communication in a single direction at a time.

(C) Communication in both directions at the same time.
(D) None of the above

24. A digital signature needs a:
(A) Private key system
(B) Shared key system
(C) Public key system
(D) All of the above

25. One way to preserve the integrity of a document is through the use of a:
(A) Thumb impression
(B) Finger print
(C) Biometric
(D) X-rays

26. Encryption and decryption provide secrecy or confidentiality but not:
(A) Authentication
(B) Integrity
(C) Key
(D) Frames

Answer Key

1. (C)	**2.** (B)	**3.** (B)	**4.** (A)	**5.** (A)	**6.** (A)	**7.** (C)	**8.** (C)	**9.** (D)	**10.** (C)
11. (B)	**12.** (C)	**13.** (C)	**14.** (C)	**15.** (D)	**16.** (C)	**17.** (C)	**18.** (A)	**19.** (C)	**20.** (A)
21. (D)	**22.** (A)	**23.** (B)	**24.** (C)	**25.** (B)	**26.** (B)				